CS675
Computer Graphics

Principles of Animation

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IIT Bombay



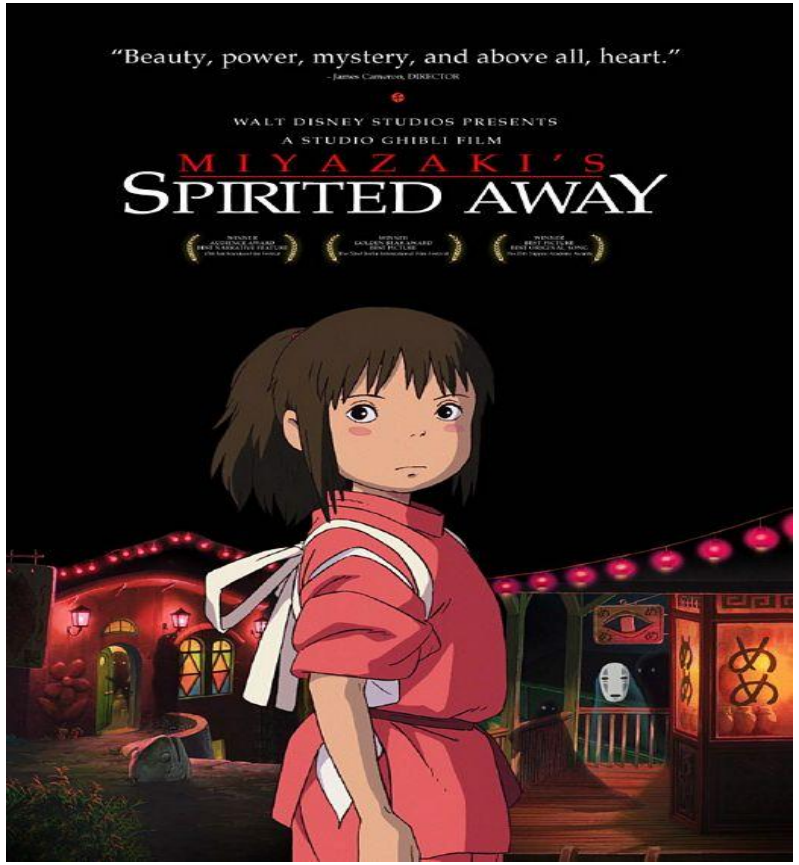
ViGIL
VISION, GRAPHICS, IMAGING & LEARNING

Animation

The term animation has a Greek (*animos*) as well as Roman (*anima*) root, meaning “to bring to life.”



Ratatouille, Pixar Animation Studios, 2007.



Spirited Away, Studio Ghibli, 2001.

Animation

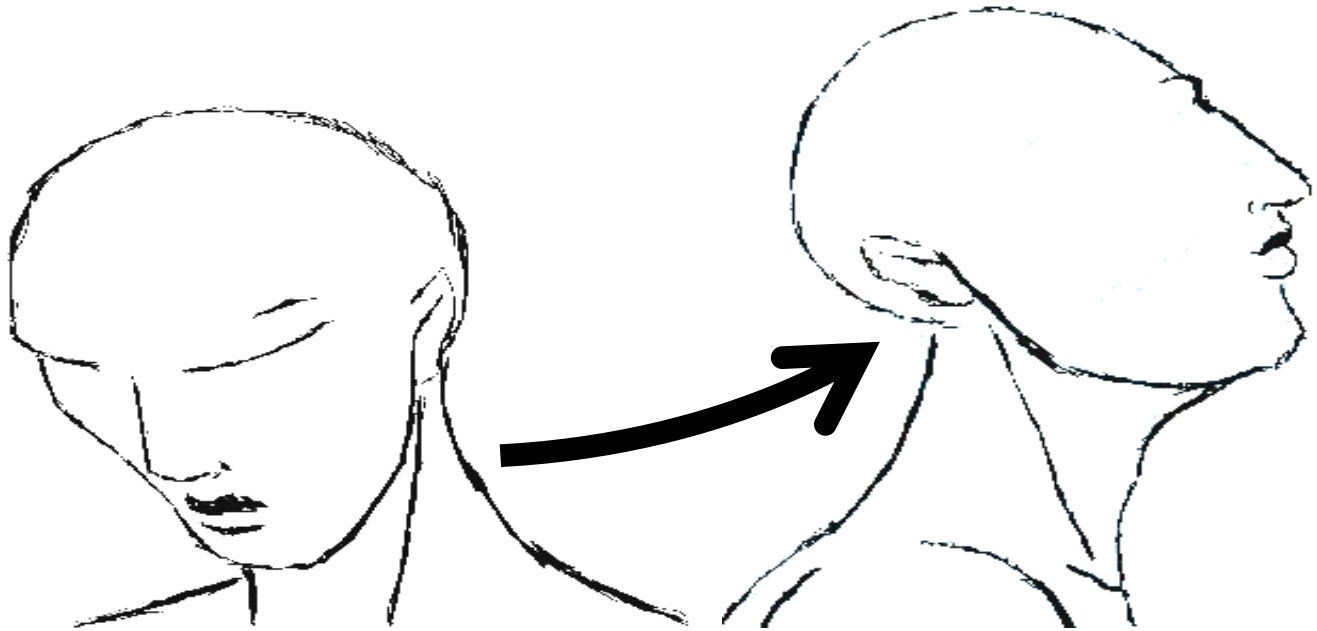
- Series of pictures of objects/characters in motion
- Displayed at some rate
  - 25 fps, 29.97 fps, 30 fps, 60fps ….
- Examples
  - Flip Books
  - Stop Motion (Claymation)
  - Traditional Hand Drawn Animation (Cel Animation)
  - Computer-assisted Keyframing
  - Motion Capture
  - Simulation

Animation

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Character Animation

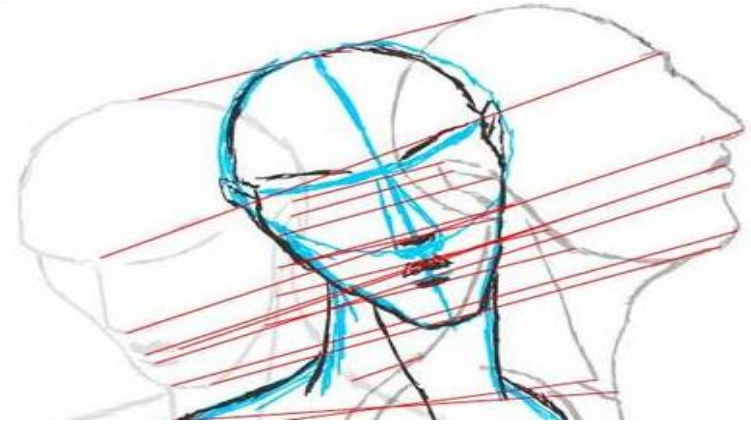
- Traditional
  - Cell Animation, hand drawn, 2D
  - Lead Animator for keyframes

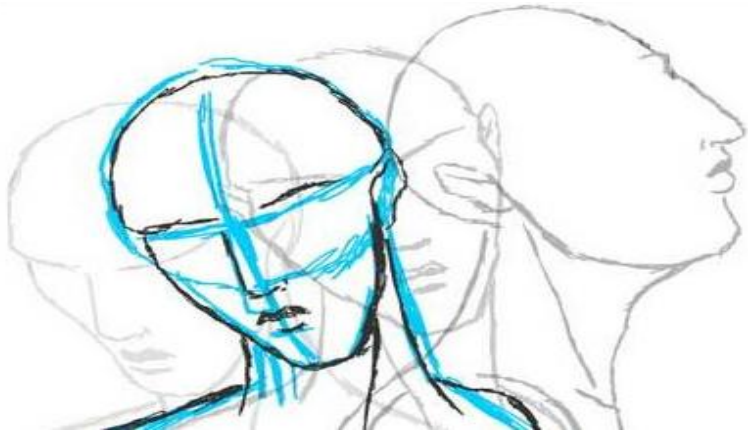


http://animation.about.com/od/flashanimationtutorials/ss/flash31detanim2.htm

Character Animation

- Traditional, hand drawn animation
  - Lead Animator for keyframes and many secondary animators for the in-betweens







http://animation.about.com/od/flashanimationtutorials/ss/flash31detanim2.htm

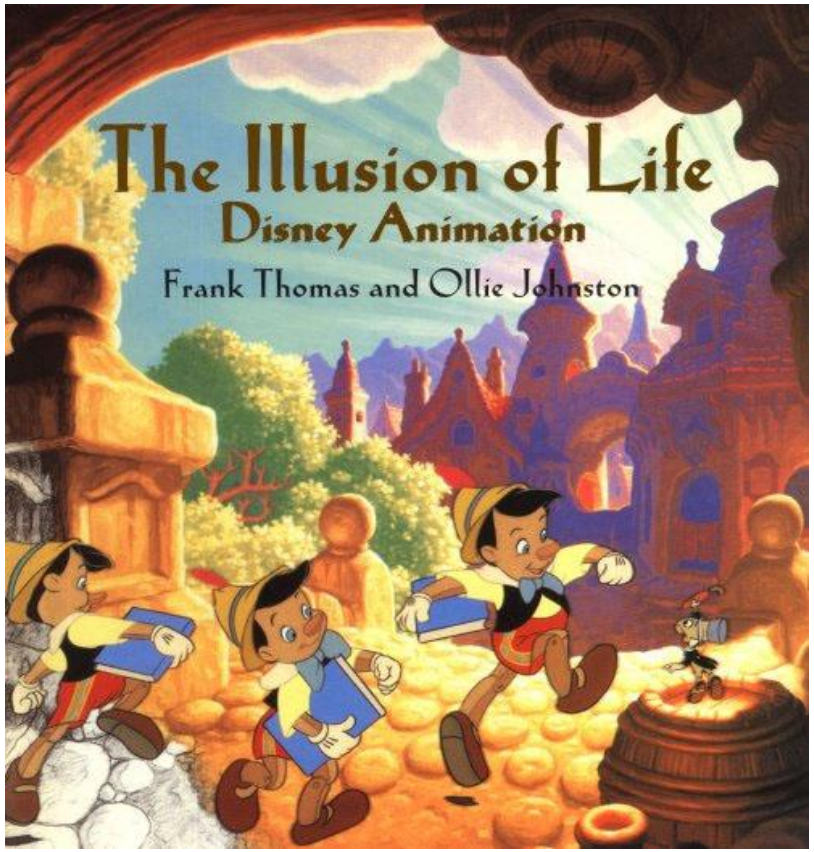
Character Animation

- Traditional, hand drawn animation
  - *Keyframing @24fps – how many drawings for a 2 hour movie?*



http://animation.about.com/od/flashanimationtutorials/ss/flash31detanim2.htm

Principles of Animation

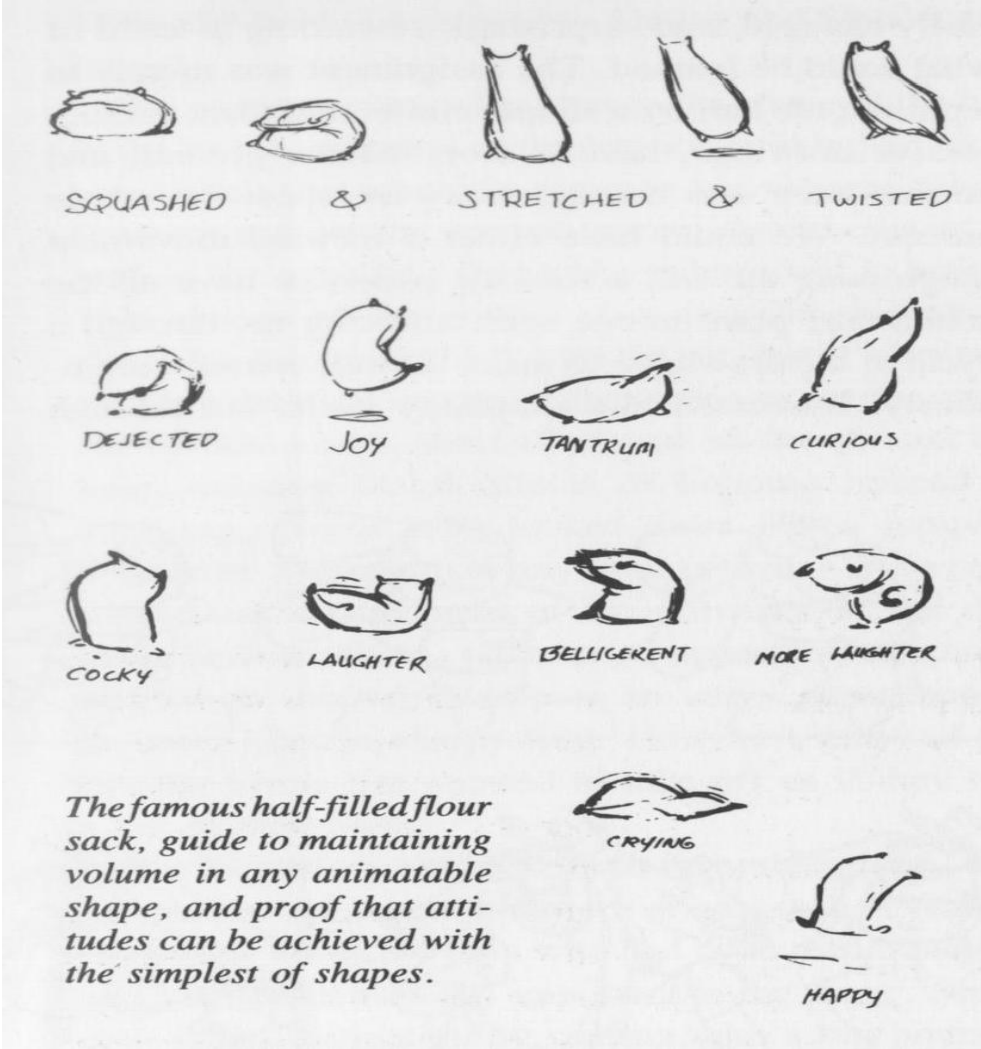


Principles of Traditional Animation Applied to 3D Computer Animation, John Lasseter, SIGGRAPH 1997.

"Illusion Of Life" by Frank Thomas & Ollie Johnston, Hyperion Press, (ISBN 0-7868-6070-7), 1981.

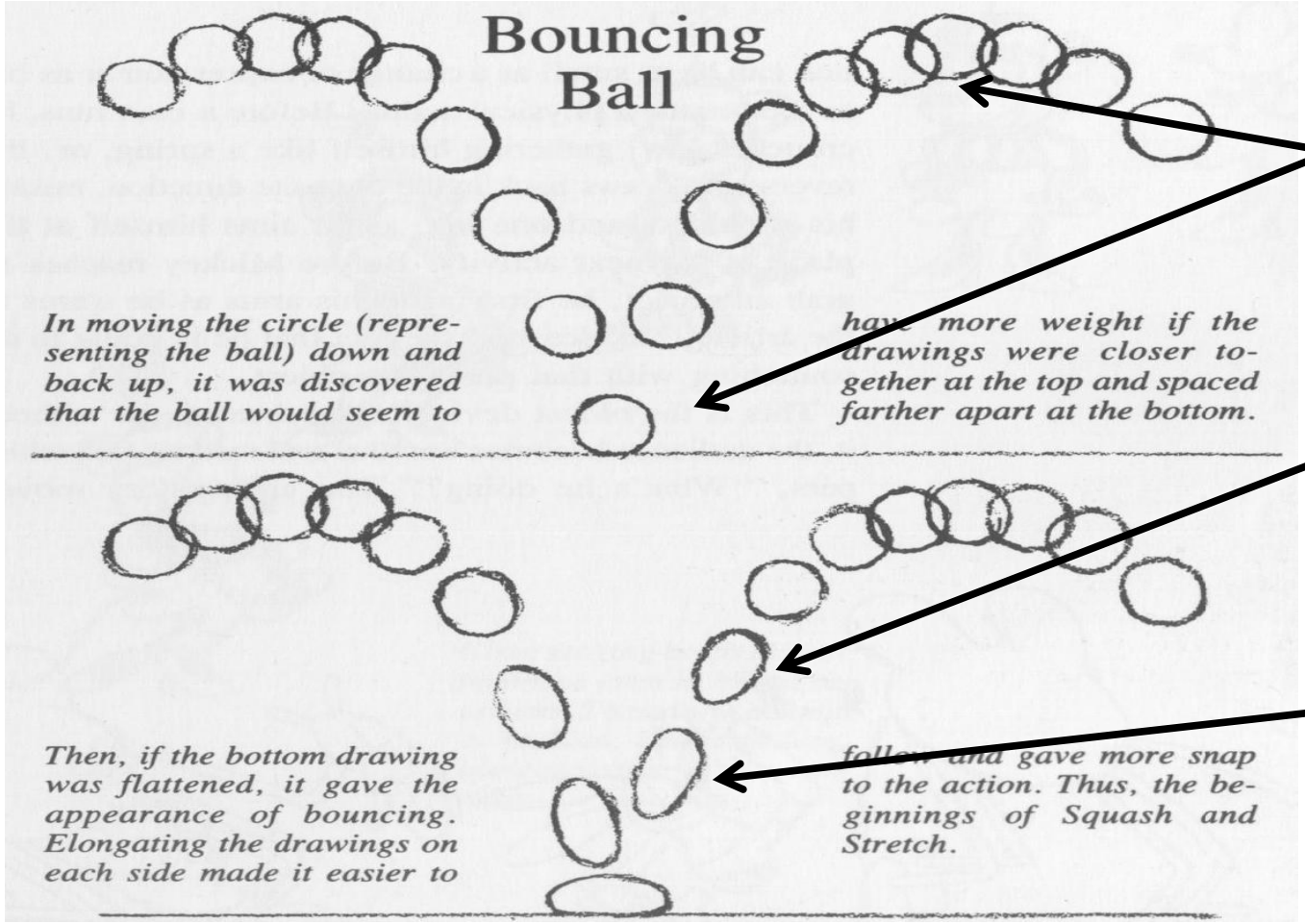
Preservation of Volume

- Characters composed of living flesh do not move rigidly (muscle flex, skin sags, etc.)
- Preservation of volume is key
- Disney animators demonstrate it here with a half-filled bag of flour.



"Illusion Of Life" by Frank Thomas & Ollie Johnston, Hyperion Press, (ISBN 0-7868-6070-7), 1981.

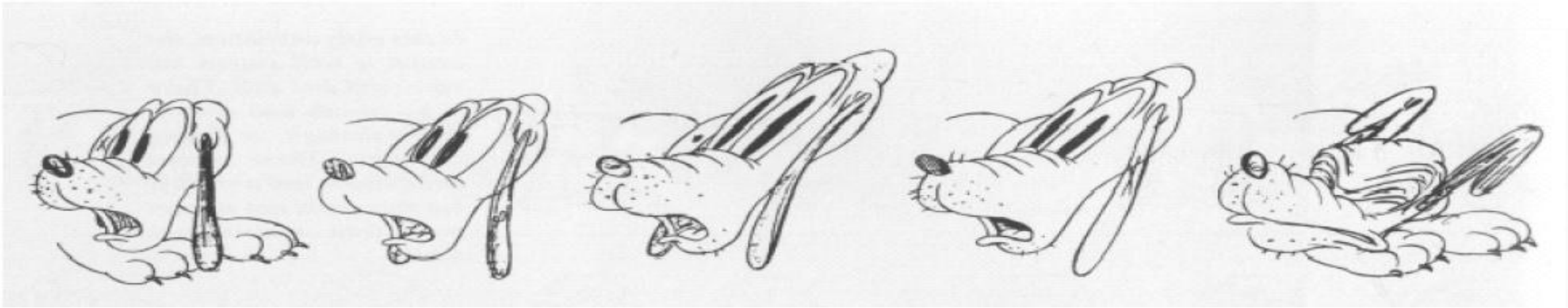
Squash and Stretch



- Weight is given to the ball using timing.
- Add squash and stretch to further define the motion and show the speed of the ball.
- A 2D representation of motion blur

"Illusion Of Life" by Frank Thomas & Ollie Johnston, Hyperion Press, (ISBN 0-7868-6070-7), 1981.

Squash and Stretch



"Illusion Of Life" by Frank Thomas & Ollie Johnston, Hyperion Press, (ISBN 0-7868-6070-7), 1981.

Approximately maintain volume...but drastically change physical characteristics over time.

Anticipation



- Tell the audience what you are going to do before you do it.
- Can be as simple as facial expressions or as broad as a body wind-up.
- What is Donald about to do?

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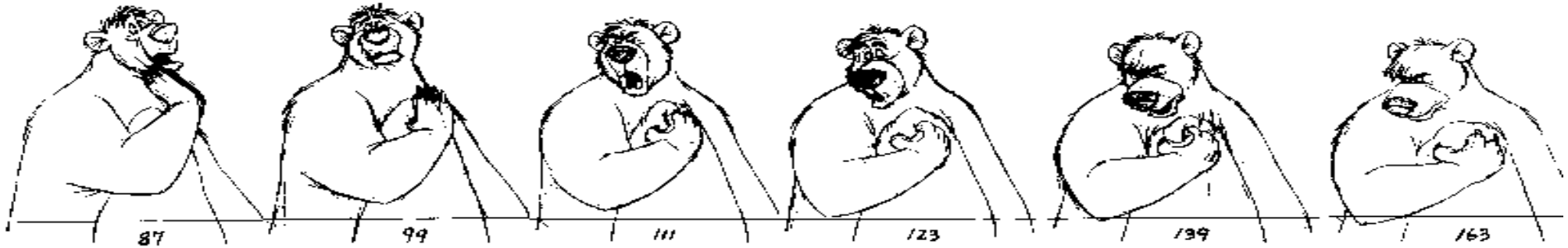
Staging



Cartoon Animation, Preston Blair, Walter Foster, 1984

- A principle borrowed from the theatre stage.
- Try to make actions to be clear in silhouette.
- Character posing and placing the camera is the key.

Straight Ahead and Pose to Pose

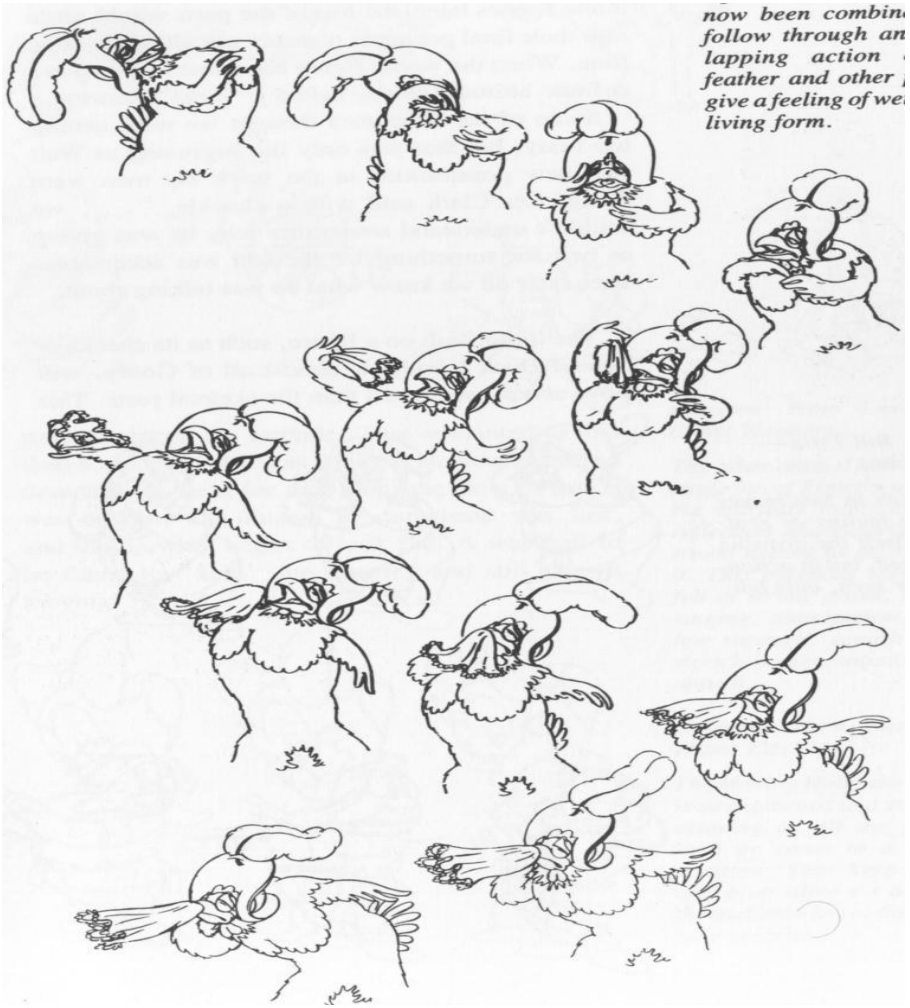


"Illusion Of Life" by Frank Thomas & Ollie Johnston, Hyperion Press, (ISBN 0-7868-6070-7), 1981.

- Two ways to animate a scene.
- Start with an idea in mind and draw all frames until done. Good for frantic motion such as a character jumping around in excitement.
- Draw the key frames first. Fill in the in-betweens.

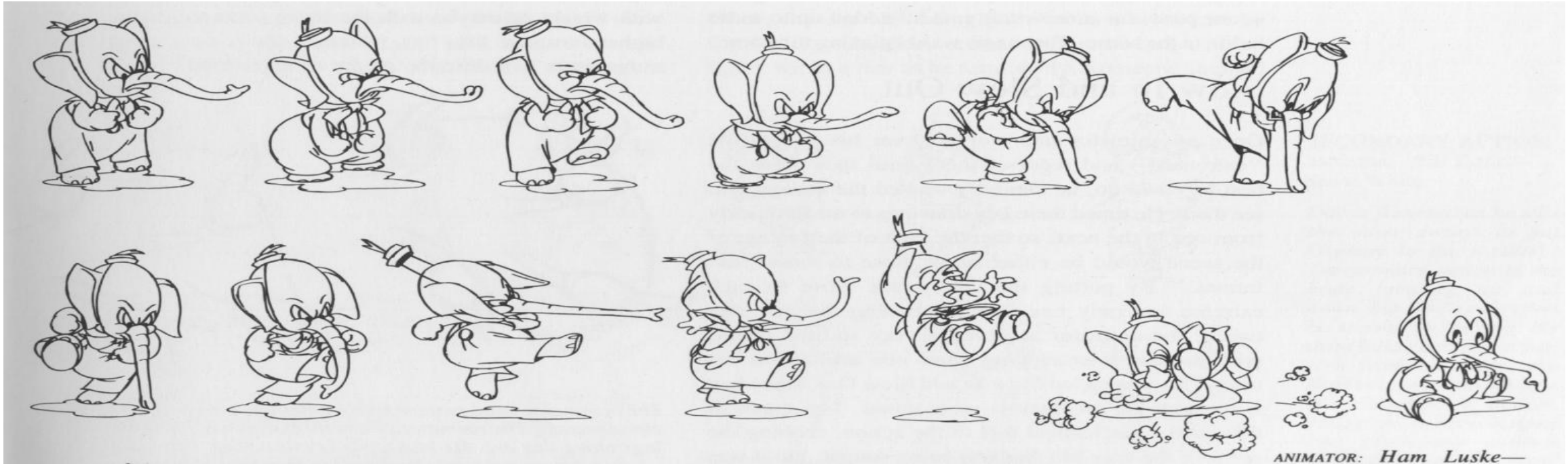
Follow Through and Overlapping Action

- Everything does not have to stop once a pose is reached.
- The way the action is completed tells us a great deal about a character. In some sense this is the opposite of anticipation... tell the audience what happened!



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Follow Through and Overlapping Action



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Slow In and Slow Out

- Animator specifies the primary or key frames which are most important.
- In order to stress these frames, move slowly away from one key frame, quickly in the in-between frames, and slowly into the next frame.
- Most time is spent on/near the key frames.







http://minyos.its.rmit.edu.au/aim/a_notes/inbetweens_01.html

Slow In and Slow Out



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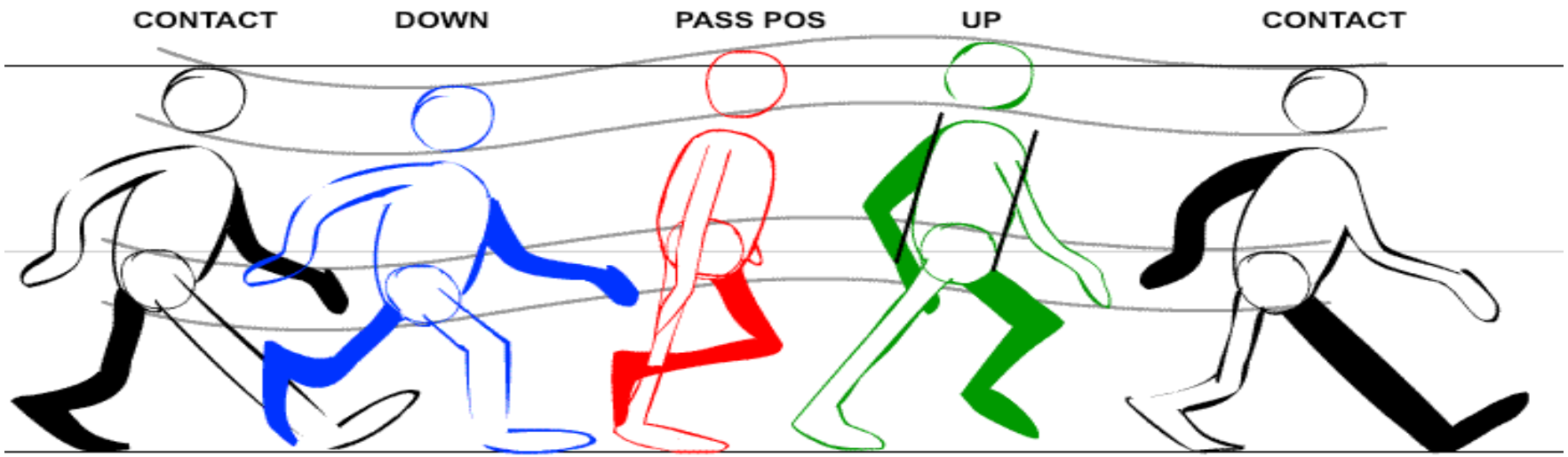
Slow In and Slow Out



http://minyos.its.rmit.edu.au/aim/a_notes/inbetweens_01.html

Arcs

- Motion in straight lines is often not *organic*.
- Most human motion happens on curved trajectories or arcs.



The Animator's Survival Kit, Richard Williams, Faber & Faber, 2002

Arcs

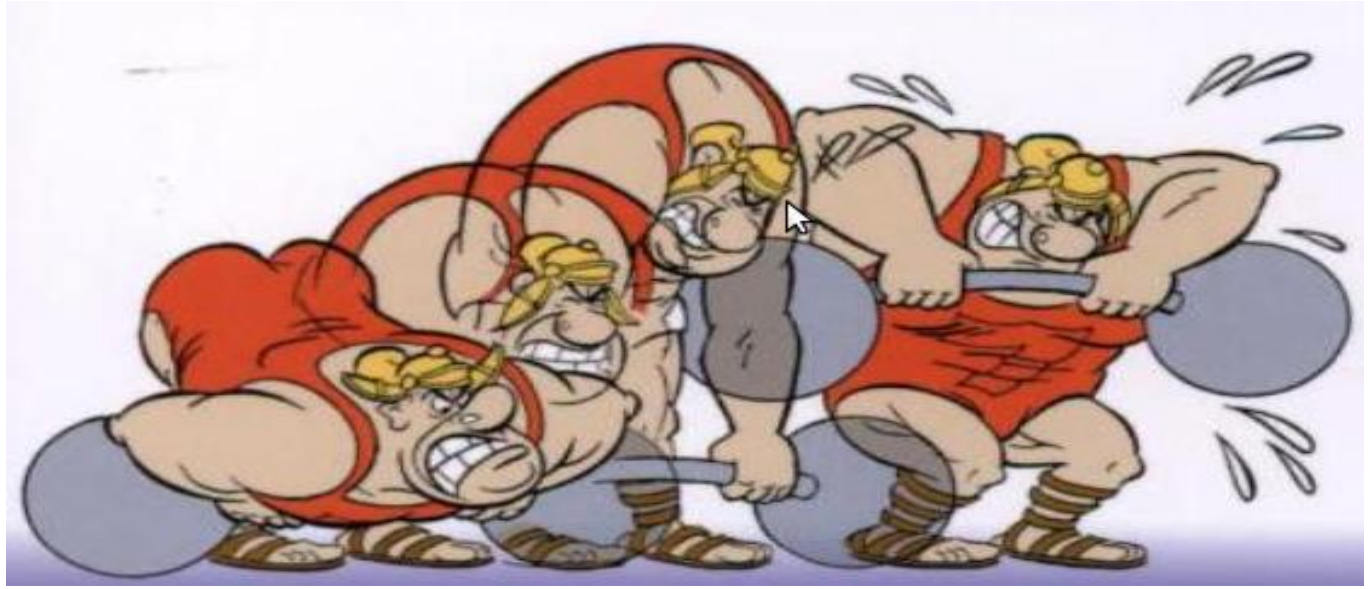
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Cartoon Animation, Preston Blair, Walter Foster, 1984

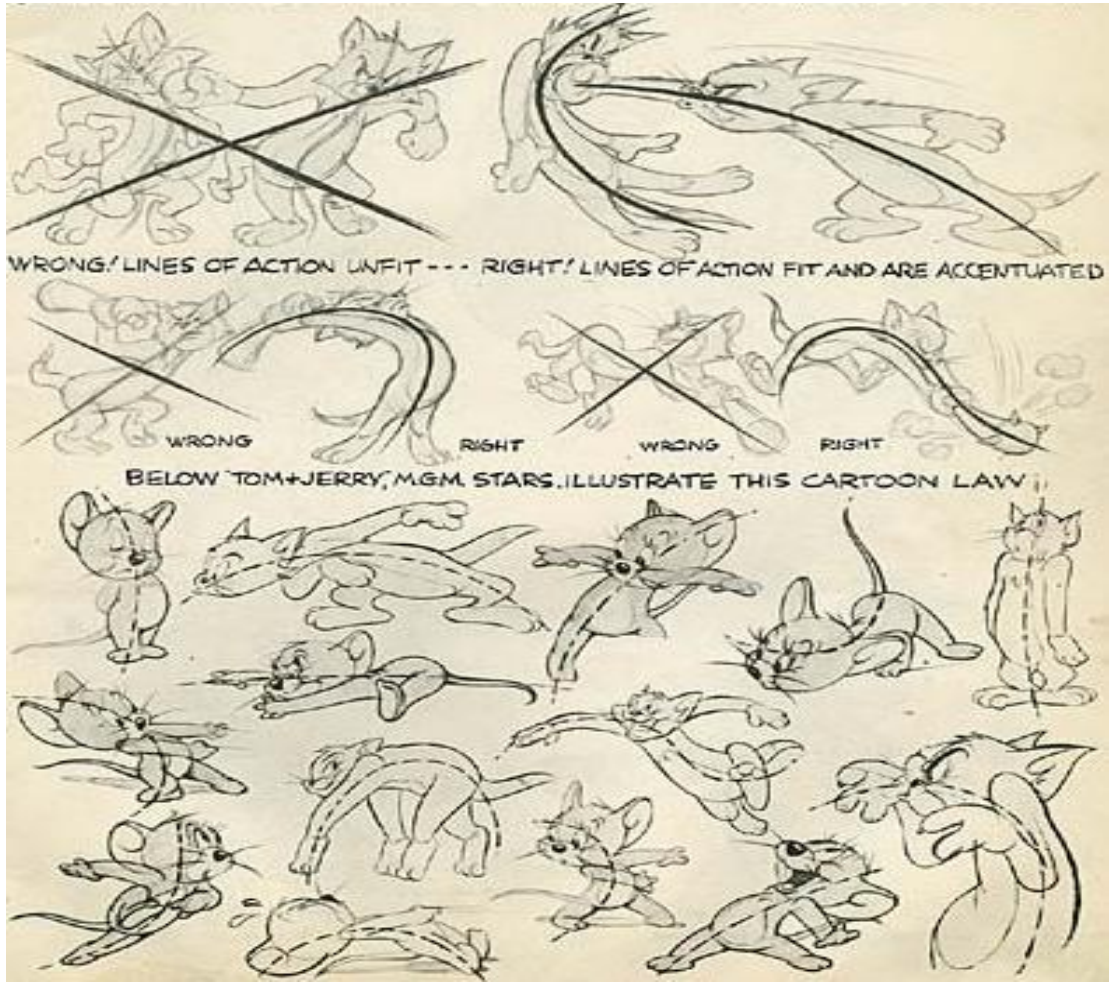
Timing

- The number of drawings determine the amount of time it takes on the screen
- no in-betweens: character hit with a huge force and his head is nearly snapped off
- two : nervous tic, muscle spasm
- five: Come on...hurry
- seven: tries to get a better look at something
- ten: stretches a sore neck



Timing for animation, Harold Whitaker, John Halas, Focal Press, 2002

Exaggeration

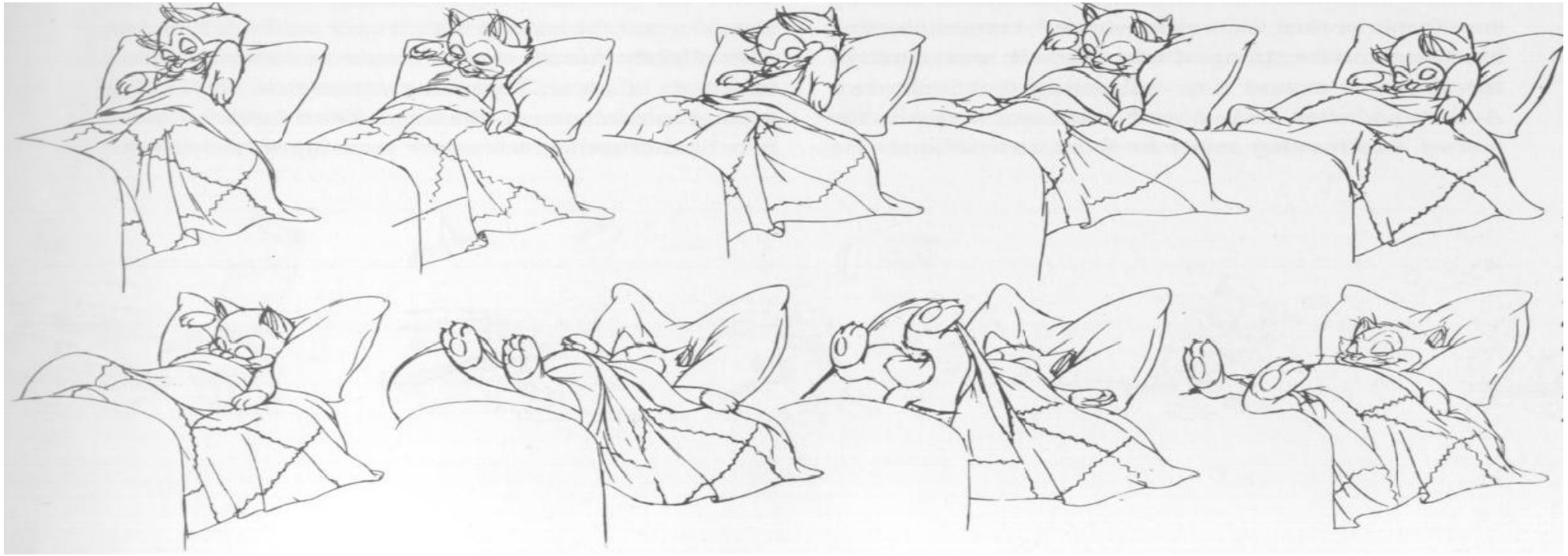


Cartoon Animation, Preston Blair, Walter Foster, 1984

- Exaggerate to make the action more believable.
- Realism and believability are difficult to achieve.
- Convey emotions.

Secondary Action

- Action aside from that of the primary character.
- Must not distract the audience.



"Illusion Of Life" by Frank Thomas & Ollie Johnston, Hyperion Press, (ISBN 0-7868-6070-7), 1981.

Solid Drawing and Appeal

- Characters are solid – have weight in the real world.
- Appeal is what makes people want to look at a character.



"Illusion Of Life" by Frank Thomas & Ollie Johnston, Hyperion Press, (ISBN 0-7868-6070-7), 1981.

How does all this lead to animation?

A Process for Animation





Beauty and the Beast, Walt Disney Feature Animation, 1991

A Process for Animation







Beauty and the Beast, Walt Disney Feature Animation, 1991

Pitch A Story

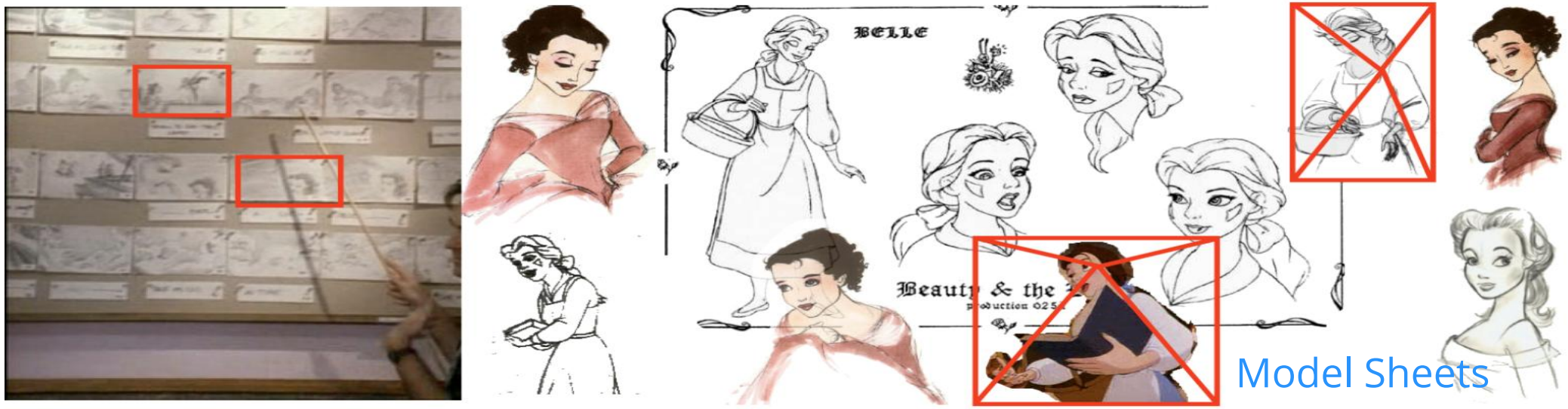






Beauty and the Beast, Walt Disney Feature Animation, 1991

Design Characters

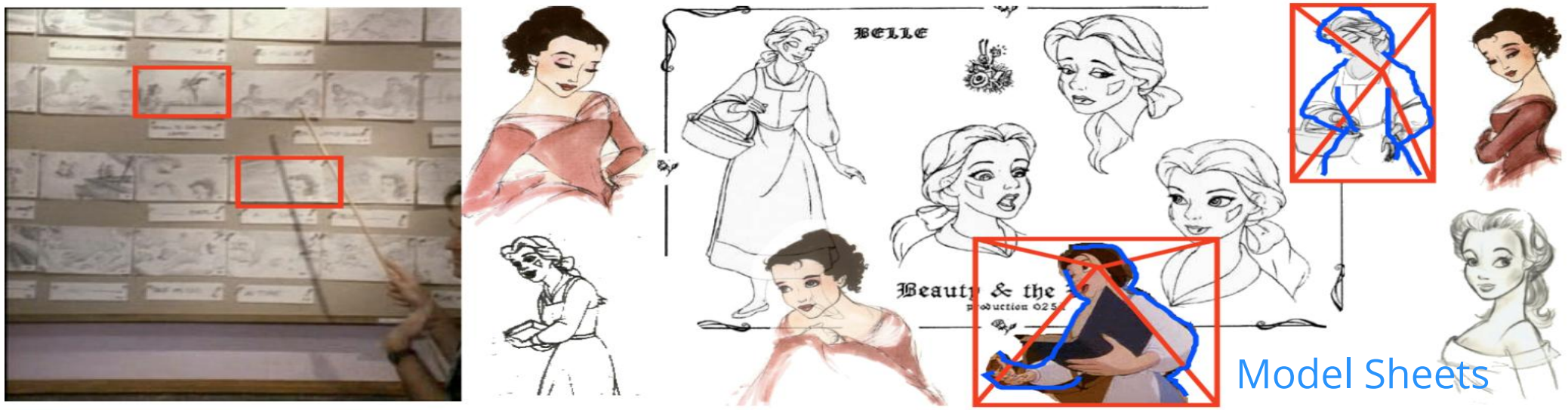


Model Sheets



Beauty and the Beast, Walt Disney Feature Animation, 1991

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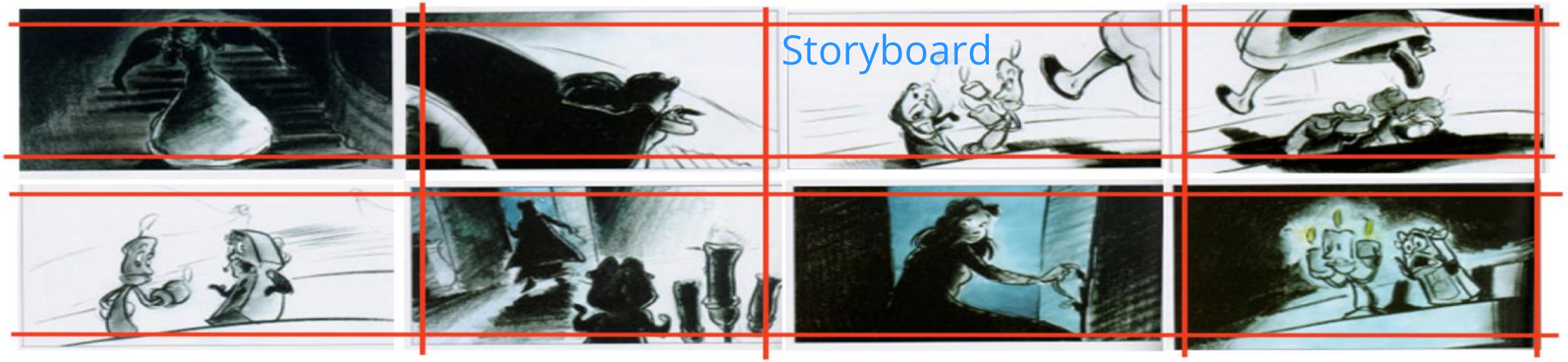
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Storyboards and Animatics

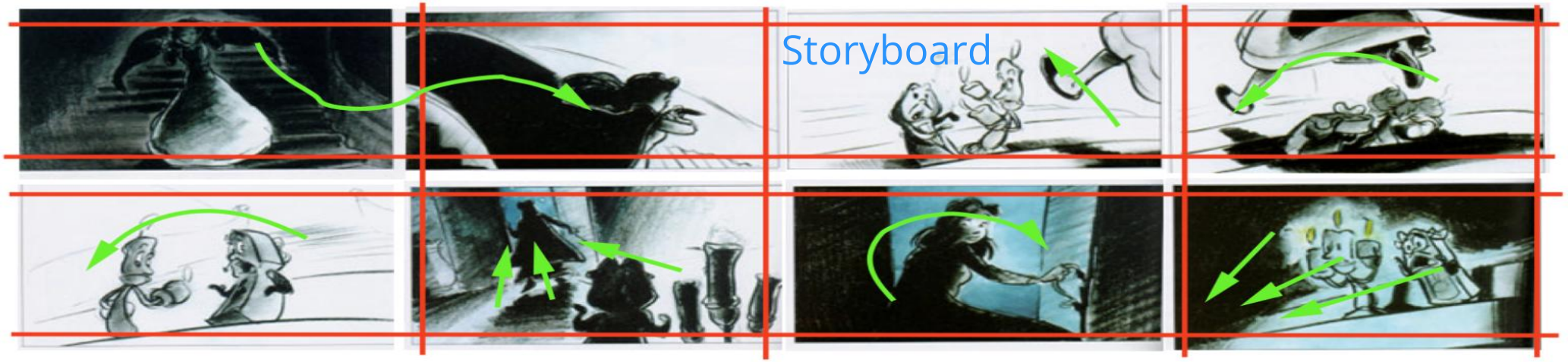




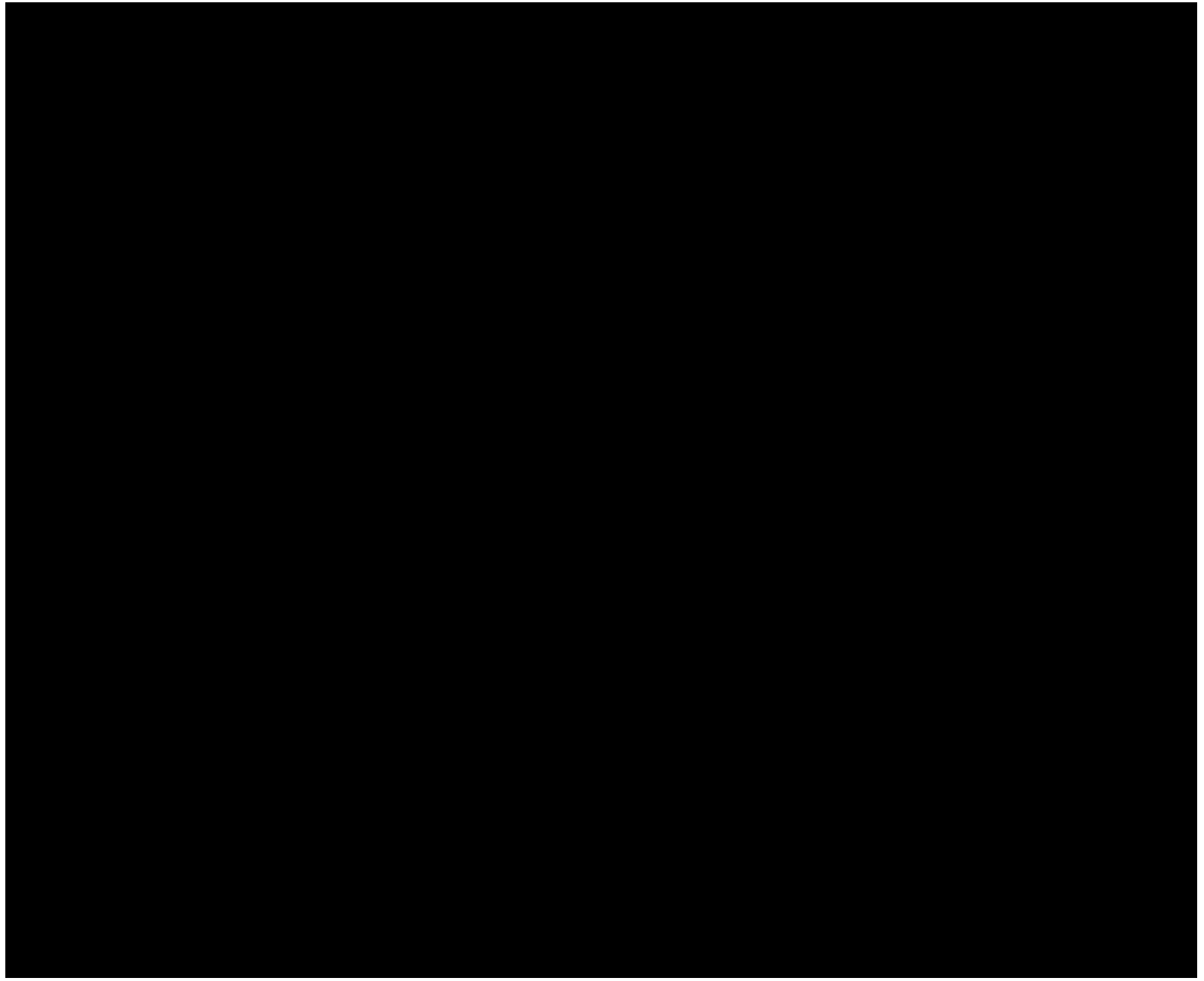
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How about 3D Characters?

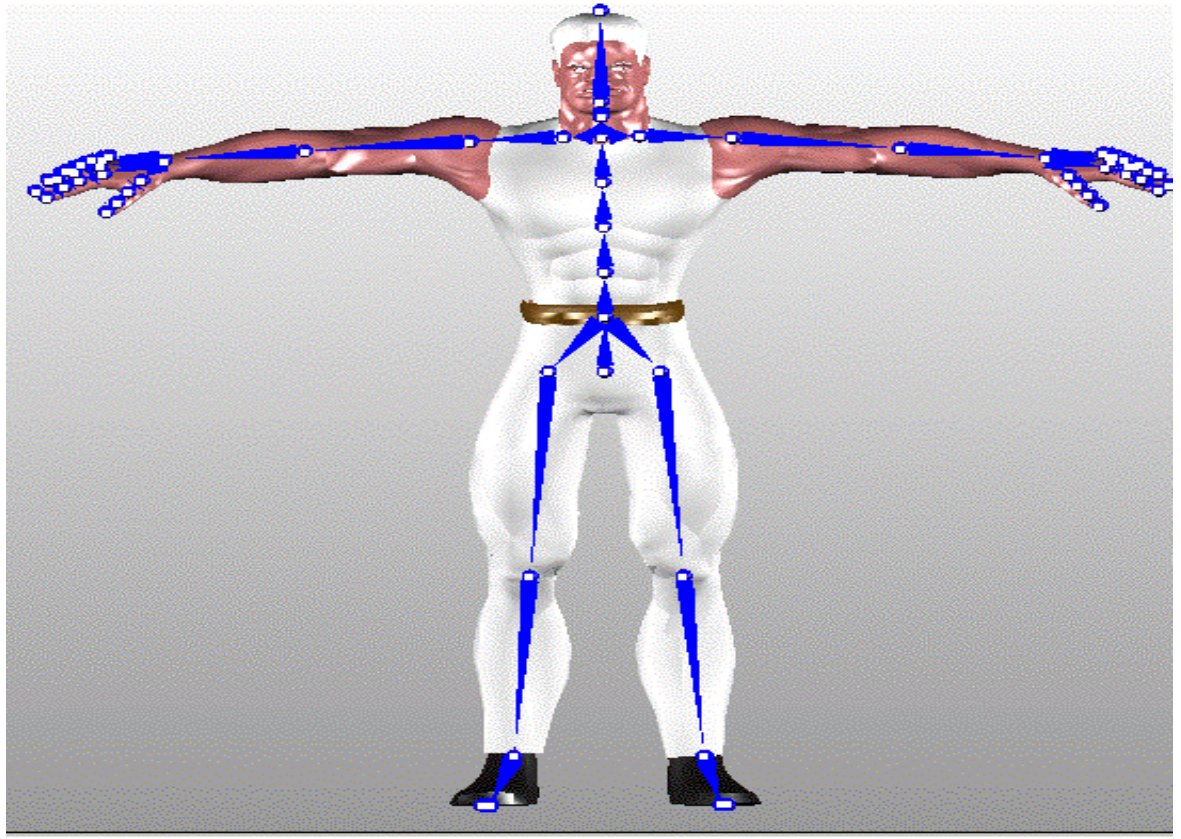
The same principles apply.



Ice Age, Blue Sky Studios, 20th Century Fox, 2002

Character Animation

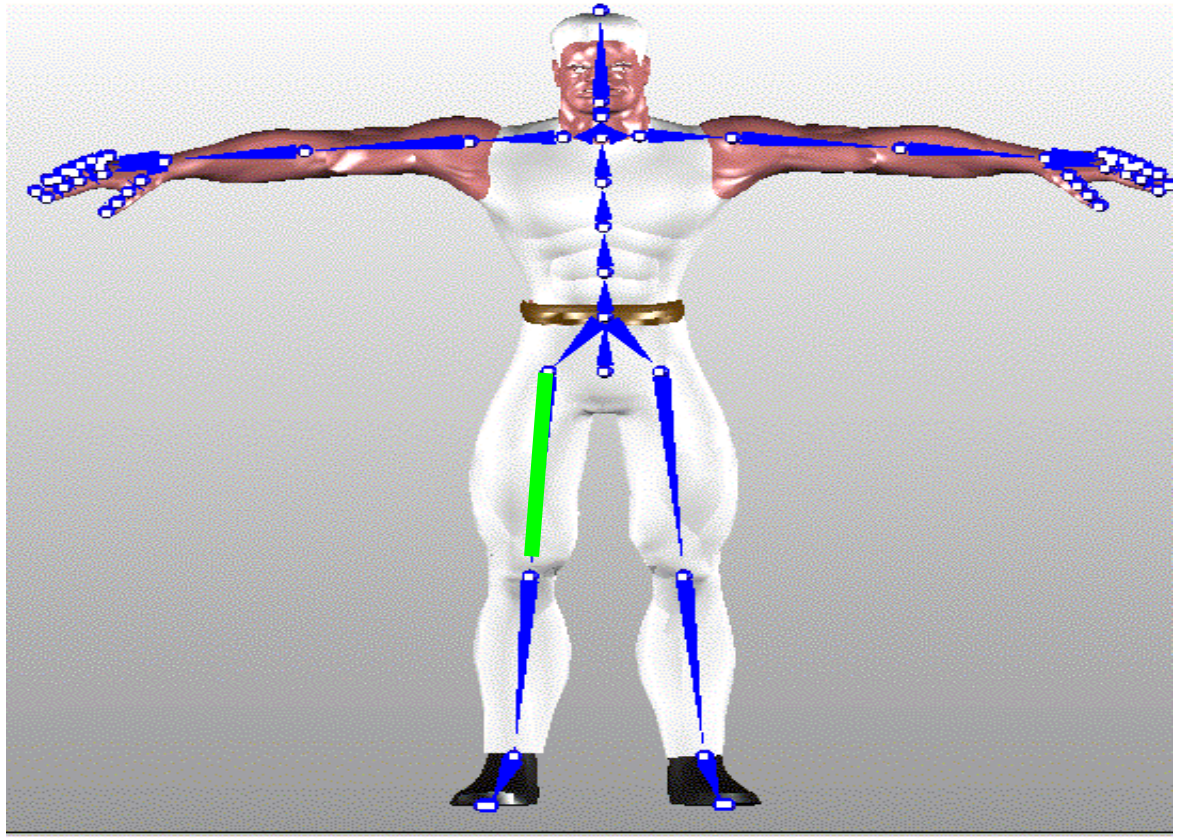
- A character in 3D is just like us.



http://www.okino.com/conv/skinning.htm

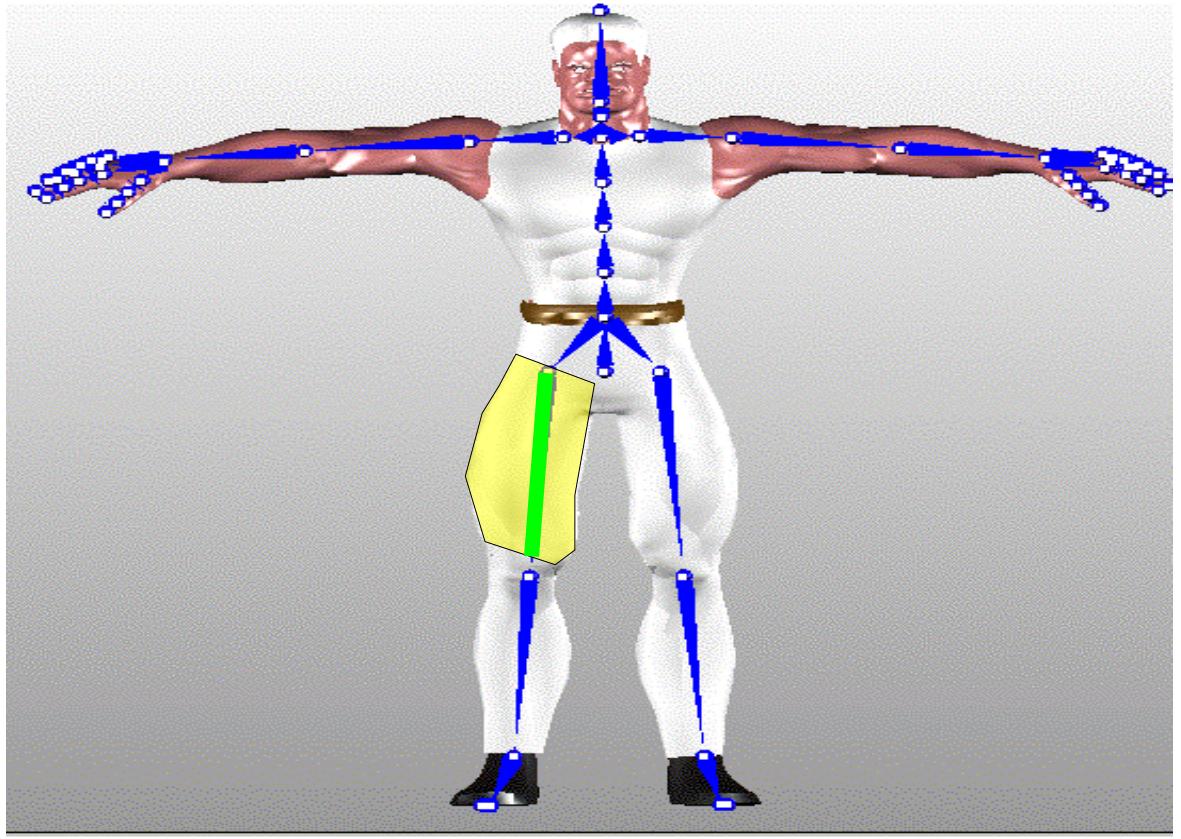
Character Animation

- A character in 3D is just like us.
- Inside they have a skeleton made up of rigid bones.



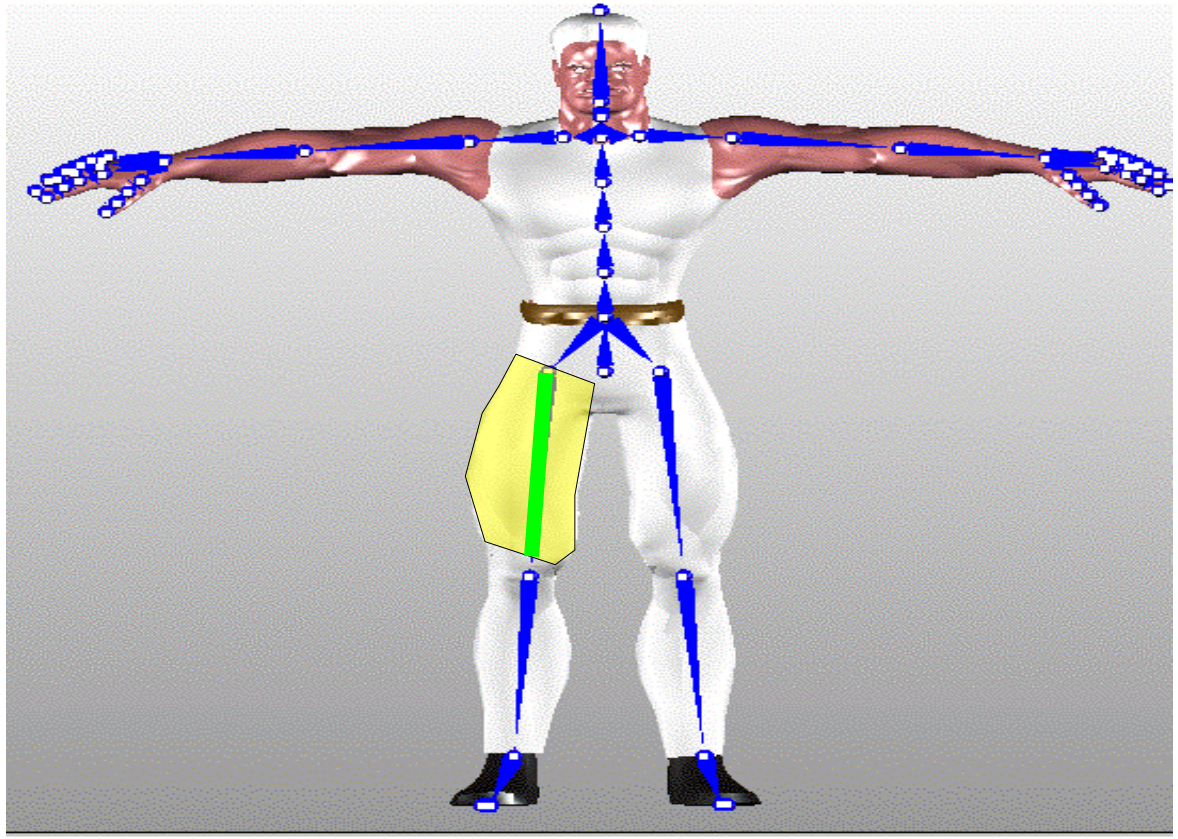
Character Animation

- A character in 3D is just like us.
- Inside they have a skeleton made up of rigid bones.
- Outside is a skin.



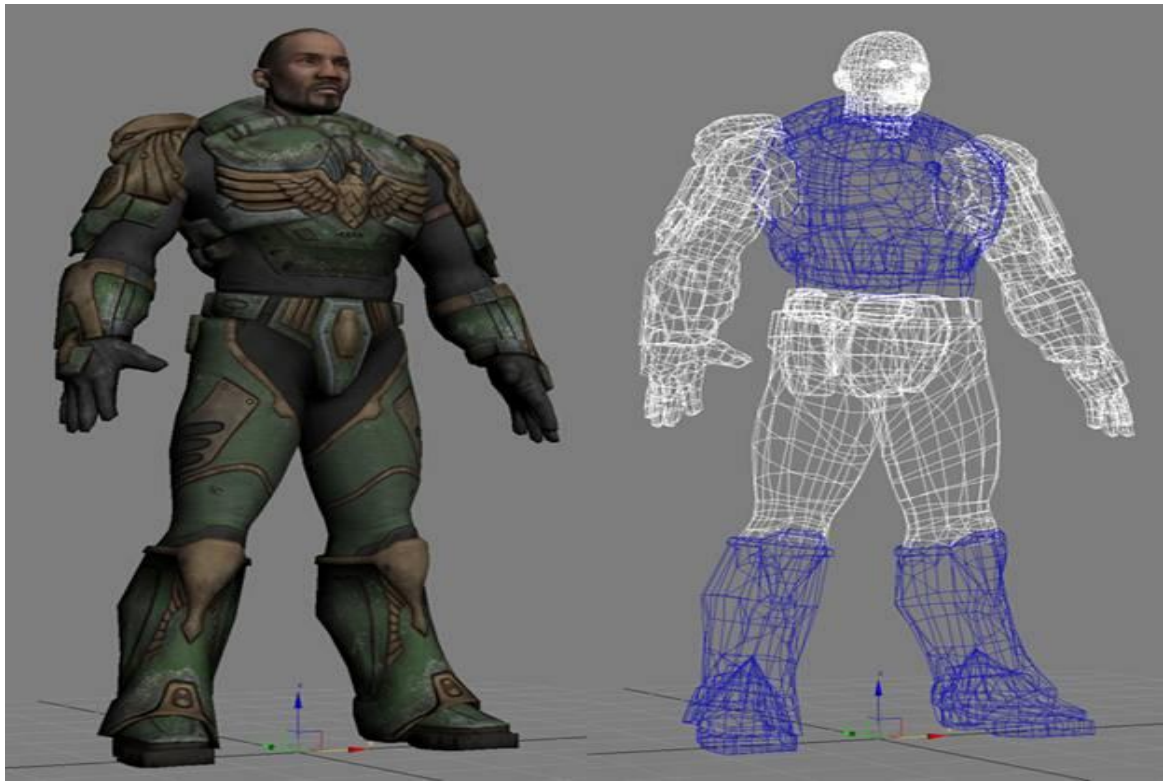
Character Animation

- A character in 3D is just like us.
- Inside they have a skeleton made up of rigid bones.
- Outside is a skin mesh.



Character Animation

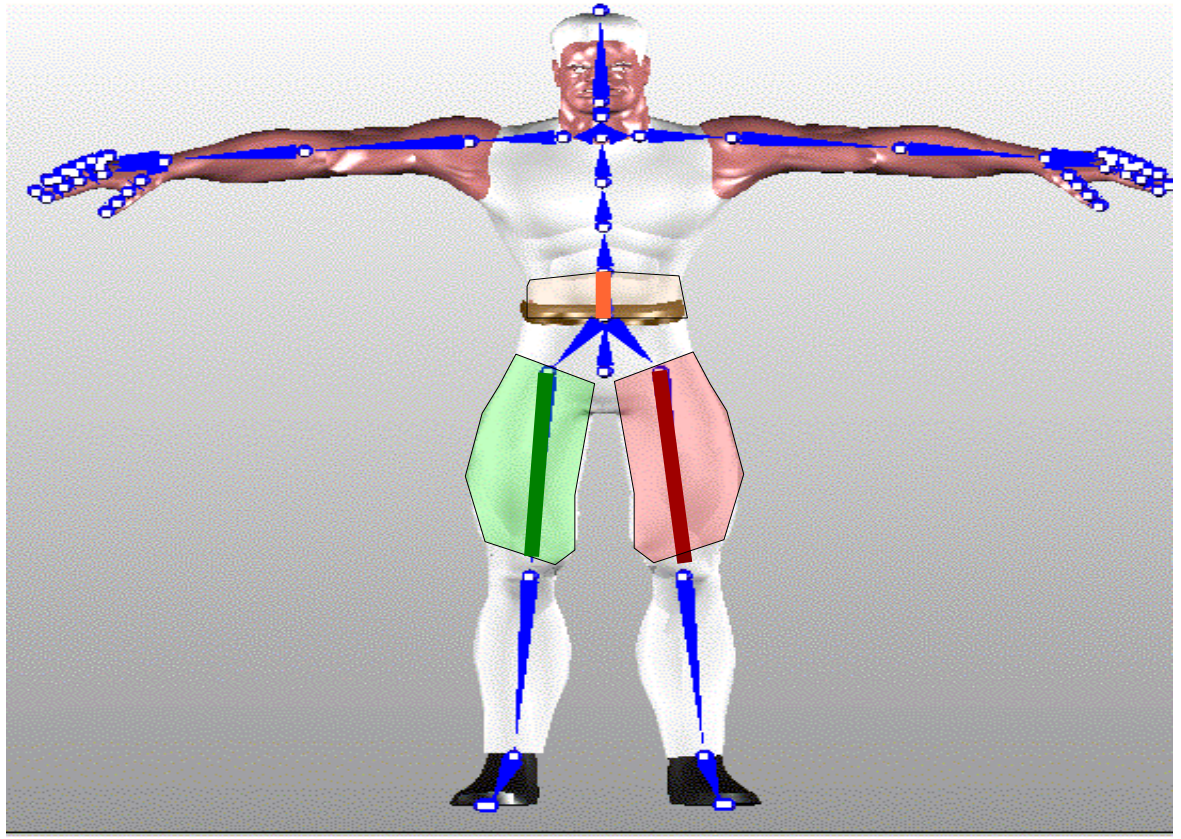
- A character in 3D is just like us.
- Inside they have a skeleton made up of rigid bones.
- Outside is a skin mesh.
- The skin mesh can be very detailed – has additional elements to capture look of the character.



http://udn.epicgames.com/Three/UT3CustomCharacters.html

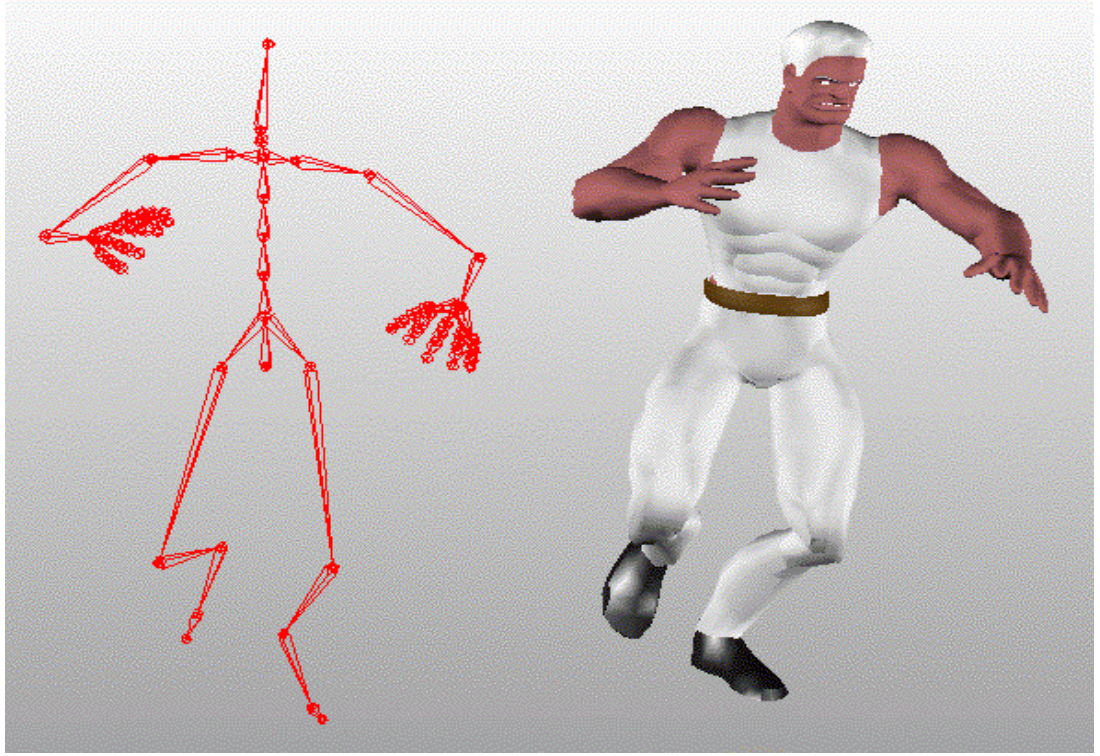
Character Animation

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- Outside is a skin mesh.
- Parts of the skin are associated to the bones.



Character Animation

- A character in 3D is just like us.
- Inside they have a skeleton made up of rigid bones.
- Outside is a skin mesh.
- Parts of the skin are associated to the bones.
- We move the bones to move the skin.



http://www.okino.com/conv/skinning.htm

Character Animation

- But how do we move the bones?
  - Manually - Keyframing
  - Mimic a performer – Motion Capture



Avatar, 2009





https://www.oktapodi.com/